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Wind power generation prediction using LSTM model optimized by sparrow search algorithm and firefly algorithm

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Abstract

As an important renewable energy source, wind power generation is highly stochastic and uncertain due to various environmental factors affecting its output. To raise the accuracy of wind power generation prediction, a bidirectional long short-term memory network combination model based on sparrow search algorithm and firefly algorithm optimization is designed. The model first employs a bidirectional long short-term memory network to capture the long-term dependency features of time series, and uses random forests for nonlinear modeling and feature selection. Then, the sparrow search algorithm and firefly algorithm are combined to optimize the hyperparameter configuration, improving the predictive performance and global search ability of the model. The findings denote that the accuracy of the designed model reaches 98.5%, with a mean square error as low as 0.005 and a prediction time as short as 0.18 s. The simulation analysis results show that the predicted values of the developed model almost coincide with the actual values, with small errors. The research outcomes denote that the optimized model greatly raises the accuracy and efficiency of wind power generation prediction, and has good application prospects.

Keywords Wind power generation, Load forecasting, Sparrow algorithm, LSTM

Introduction

In recent years, due to the escalating global energy crisis and environmental pollution issues, promoting the advancement and utilization of renewable energy has emerged as a shared objective of the international community. Wind power has been widely adopted and rapidly developed worldwide as a clean and efficient renewable energy source [1]. Compared to conventional fossil fuels, wind energy offers significant environmental advantages and can effectively mitigate greenhouse gas emissions and atmospheric pollutants [2]. However, the output power of Wind Power Generation (WPG) is highly variable and susceptible to fluctuations, primarily influenced by a range of unpredictable factors, including meteorological conditions, and variations in wind speed and direction. These factors present considerable challenges to the secure and reliable operation of the power system following the integration of wind power into the grid. Currently,

wind power forecasting methods are primarily categorized into two types: physical models and data-driven models. The physical model establishes a mathematical framework for prediction based on meteorological conditions, terrain features, and other factors. However, it exhibits strong dependence on environmental conditions and performs inadequately in complex environments. In contrast, data-driven models based on deep learning and machine learning have increasingly become the dominant approach for WPG prediction due to their robust data mining and modeling capabilities. Tang J et al. developed a prediction method based on variational mode decomposition and Bidirectional Long Short-Term Memory (BiLSTM). The findings indicated that this method effectively enhanced the prediction Accuracy (ACC) and stability by decomposing the load sequence and optimizing hyperparameters, thereby more accurately tracking the trend of load changes [3]. To develop an efficient and accurate short-term power load forecasting method, Ciechulski et al. developed a prediction model based on recursive Long Short-Term Memory (LSTM) and applied it to 24-hour load forecasting within the Polish power system. The research results demonstrated that the model could effectively predict irregular trends, and the ensemble prediction method significantly enhanced the prediction ACC compared to a single predictor, reducing errors by more than 6% [4]. To accurately predict short-term power load, enhance power grid decision-making, and improve user power management, Liu et al. developed a prediction model based on long-term and short-term time series networks. The findings indicated that the model demonstrated good ACC and stability in load forecasting by analyzing the correlation between variables and load, in conjunction with convolutional neural networks (CNN) and LSTM networks [5].

Bao et al. proposed a hybrid prediction model based on empirical mode decomposition and Deep Neural Networks (DNNs), aiming to enhance the ACC of short-term electricity predictions. The model first applied empirical mode decomposition for denoising, followed by inputting the processed data into a DNN comprising CNN and LSTM for feature analysis and prediction. The findings indicated that the power prediction performance of this model was superior, outperforming other comparative models [6]. Abbasi et al. developed an architecture that integrates multi-condition scheduling algorithms with multi-level queues to optimize workflow scheduling for Internet of Things (IoT) devices in fog/cloud environments. The architecture employed CPE to calculate priorities and LSTM to predict workloads. The research results demonstrated that this method significantly enhanced system throughput and efficiency, with throughput increasing by 48.6% and communication costs decreasing by 56%. Additionally, significant improvements were observed in production span, waiting time, and parallelism [7]. Xiao et al. developed a multi-channel metal foam latent heat storage tank design and constructed a three-dimensional numerical model to optimize the performance of the latent heat storage system. Additionally, an LSTM-BP neural network prediction model was developed. The findings indicated that increasing the velocity or temperature of the heat transfer fluid could shorten the melting time of phase change materials, and machine learning models offered an adaptive new method for thermal storage design [8]. Zhang et al. developed a prediction model based on stacked BiLSTM to accurately predict the remaining lifespan of supercapacitors. The research results demonstrated that the model performed well when the number of hidden layers was set to 2, with

prediction errors Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) of 0.0275 and 0.0241, demonstrating excellent performance [9]. (Table 1)

In summary, numerous scholars have employed intelligent optimization algorithms, such as BiLSTM, in WPG prediction. However, most studies rely on a singular optimization strategy and do not fully address the challenges of hyperparameter selection and local optima in the model. Consequently, an LSTM model for WPG prediction, optimized using the Sparrow Search Algorithm (SSA), is proposed. The model initially employs a BiLSTM network to capture the long-term dependencies of time series data and utilizes the Random Forest (RF) algorithm for nonlinear modeling and feature selection. Subsequently, the SSA is applied to jointly optimize the hyperparameter configuration, enhancing the predictive performance and global search capability of the model. The novelty of this research lies in combining the Firefly Algorithm (FA) with SSA to optimize the hyperparameter configuration of BiLSTM and RF, thereby improving model performance. The study aims to provide a reliable prediction method for WPG. A BiLSTM-RF combination model based on FA-SSA optimization has been proposed, which is innovative in hyperparameter optimization, model fusion, and computational efficiency. By adopting the FA-SSA joint optimization strategy and combining the global search capability of the FA with the local optimization capability of the SSA, the efficiency of hyperparameter optimization is improved and convergence is accelerated. By combining BiLSTM with RF, BiLSTM is used for time series modeling, while RF is used for feature selection and nonlinear modeling, enhancing the predictive ability of the model while reducing computational complexity. RF technology is used to filter key functions, reduce redundant data entry and optimize computing resources.

Table 1 Literature review

Author	Methodology	Key contributions	Limitations
Tang J et al. [3]	Variational Mode Decomposition + BiLSTM	Enhanced prediction accuracy and stability by decomposing the load sequence and optimizing hyperparameters	Dependent on load sequence and may not generalize to all conditions
Ciechulski et al. [4]	Recursive LSTM	Developed a model for 24-hour load forecasting, improving prediction accuracy by over 6%	Limited to Polish power system; may not apply universally
Liu et al. [5]	Long-term and short-term time series networks (LSTM + CNN)	Good accuracy and stability in load forecasting by analyzing correlations between variables	Complex model architecture; may not be optimal for all forecasting task
Bao et al. [6]	Empirical Mode Decomposition + Deep Neural Networks (CNN + LSTM)	Denosing with empirical mode decomposition, followed by deep neural networks for prediction	Focus on short-term electricity; may require high computational cost
Abbasi et al. [7]	Multi-condition scheduling with LSTM	Optimized workflow scheduling for IoT devices, enhancing throughput and efficiency	Not focused on WPG; more applicable to IoT applications
Xiao et al. [8]	LSTM-BP neural network + metal foam latent heat storage tank design	Optimized thermal storage design using machine learning models	Application to thermal storage; limited relevance to energy forecasting
Zhang et al. [9]	Stacked BiLSTM	Accurately predicted the remaining lifespan of supercapacitors with low RMSE and MAE	Focuses on supercapacitors, not on WPG

Methods

The first section proposes a wind power forecasting model grounded on BiLSTM-RF to solve the ACC issue of WPG. In the second section, SAA is introduced to improve the model for the problem of excessive and difficult to determine hyperparameters.

Wind power generation prediction model based on BiLSTM-RF

As a clean and renewable energy source, WPG has significant uncertainty and randomness in its output, which is influenced by various factors such as weather, wind speed, and wind direction [10, 11]. Accurate WPG forecasting can help optimize power grid scheduling, reduce backup capacity, lower power generation costs, and is of great significance for electricity market trading and economic benefit evaluation of wind farms. A WPG prediction model based on BiLSTM is developed. The BiLSTM network is composed of two LSTMs with opposite flow directions, and its structure is shown in Fig. 1.

In Fig. 1, BiLSTM is an improved LSTM network that can simultaneously consider both past and future information of time series data. The figure is divided into a forward LSTM layer at the top and a reverse LSTM layer at the bottom. The forward LSTM transmits information in chronological order from x_t to x_{t+n} , while the reverse LSTM processes information in reverse order from x_{t+n} to x_t . At each moment, the input x_t is simultaneously fed into the forward and backward LSTM layers, which calculate the corresponding hidden states h_t and h'_t , respectively, i.e. forward hidden state and reverse hidden state [12, 13]. Subsequently, these two hidden states are fused as the final output of the current time y_t . The structure of BiLSTM is similar to LSTM, consisting of a forgetting gate, inputting gate, and outputting gate. The expression for the forgetting gate is indicated in Eq. (1).

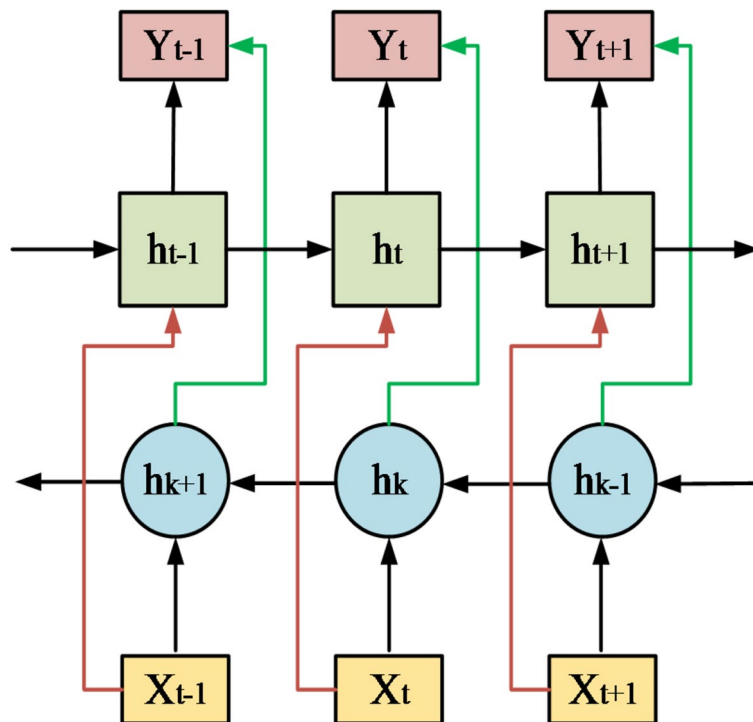


Fig. 1 BiLSTM network structure

$$f_t = \sigma(w_f[h_{t-1}, x_t] + b_f) \quad (1)$$

In Eq. (1), x_t represents the input, h_{t-1} means the previous hidden state, σ represents the activation function, b means the threshold matrix, and w represents the weight matrix. The inputting gate determines whether the current input information is added to the cell state, and its expression is shown in Eq. (2).

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (2)$$

In Eq. (2), i_t represents the inputting gate, W_i denotes the weight matrix, and b_i denotes the threshold matrix. The outputting gate controls whether the information in the current cell state is transmitted to the hidden state, as expressed in Eq. (3).

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (3)$$

In Eq. (3), o_t represents the outputting gate, W_o denotes the weight matrix, and b_o means the threshold matrix. Although BiLSTM excels at modeling long-term dependencies and temporal features of time series data, a single BiLSTM network has limitations when dealing with nonlinear relationships and high-dimensional features [14]. Therefore, the study introduces RF to improve it. RF is based on a set of decision trees, using Bagging techniques and feature random selection strategies. RF improves the predictive performance and robustness of the model by integrating a large number of weak classifiers. The RF algorithm structure is indicated in Fig. 2.

In Fig. 2, firstly, the original dataset is sampled using Bootstrap sampling technique to randomly extract multiple sub sample sets with replacement, and each sub sample set is utilized to train a decision tree. Each decision tree randomly selects some features for splitting during training, forming multiple independent decision tree models. The decision tree structure is represented by hierarchical nodes, where each node is a feature splitting point [15]. Once the training period has concluded, the next stage is the integration phase. The final result is then decided through a voting process. For classification

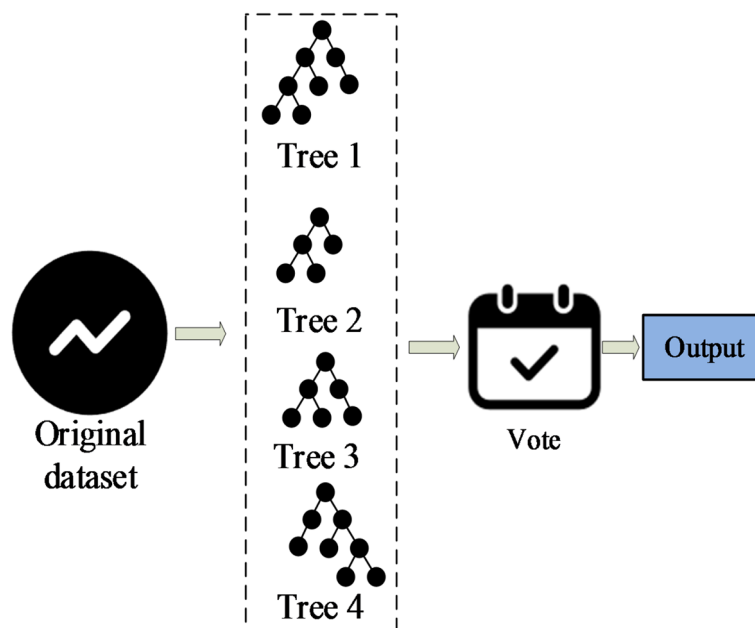


Fig. 2 RF structure diagram

tasks, RF will use majority voting to determine the final category based on the prediction results of all decision trees. For regression tasks, the average output value of all decision trees is calculated as the final prediction result. Each decision tree in RF uses the Gini index as a criterion when splitting nodes, and its expression is shown in Eq. (4).

$$Gini(D) = 1 - \sum_{i=1}^C p_i^2 \quad (4)$$

In Eq. (4), D refers to the dataset of the current node, C stands for the number of categories, p_i represents the proportion of samples in category i , and when splitting nodes, features and thresholds that minimize the Gini index are selected. The information gain of RF is shown in Eq. (5).

$$IG(D, A) = H(D) - \sum_{v \in V} \frac{|D_v|}{|D|} H(D_v) \quad (5)$$

In Eq. (5), $H(D)$ represents the entropy of dataset D , A represents the features used for splitting, V represents possible values, and D_v represents subsets. RF is used to optimize BiLSTM, and its model structure is shown in Fig. 3.

In Fig. 3, firstly, in the initial stage, raw data is collected and preprocessed, including steps such as missing value processing and normalization. After data preprocessing, it is broken into a training set and a testing set. In the training section, the hyperparameters of the BiLSTM-RF network is defined, including the amount of nodes in the BiLSTM layer, learning rate, etc., and then the loss function is calculated, such as Mean Square Error (MSE), to guide model training. After training, the BiLSTM-RF network model can be obtained and saved for prediction purposes. In the testing section, it loads the trained model and performs load forecasting on the test set. The predicted results are subjected to inverse normalization and restored to their original dimensions. Finally, the regression error and evaluation metrics are calculated to evaluate the efficacy of the model. The final Predicted Value (PV) is output and the entire process is completed.

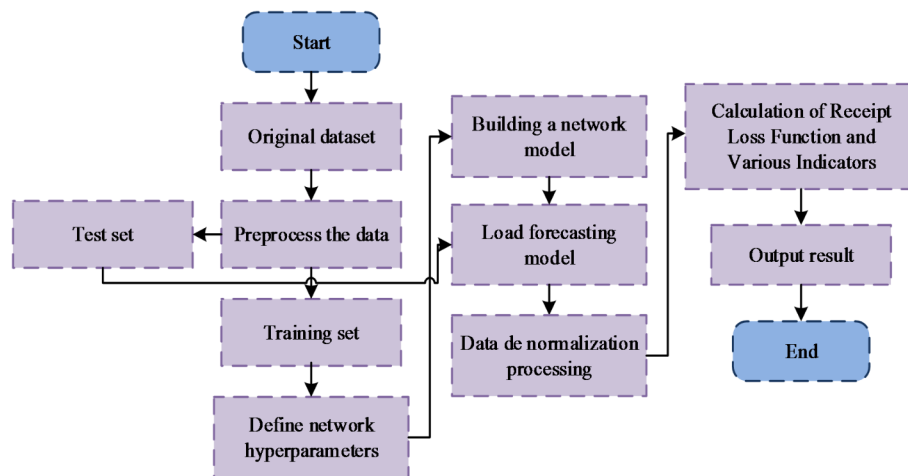


Fig. 3 BiLSTM-RF model structure

Optimized LSTM model for wind power generation prediction based on SSA

Although BiLSTM excels at handling the long short-term dependencies of time series data, RF is used for feature selection and nonlinear modeling. However, the BiLSTM-RF model also has some shortcomings. Both BiLSTM and RF contain a large number of hyperparameters, such as the amount of hidden layers, nodes, learning rate of BiLSTM, as well as the number of trees and feature selection of RF [16]. These hyperparameters have a critical influence on model performance, and traditional manual parameter tuning or grid search methods are inefficient and prone to getting stuck in local optima. Therefore, SSA is used for parameter selection in the study. SSA simulates the ecological behavior of sparrows foraging and uses a group of individual sparrows as candidate solutions for iterative optimization. Each sparrow individual represents a solution, and they move in the search space to find the optimal solution. Its structure is shown in Fig. 4.

In Fig. 4, it randomly generates sparrow individuals and initializes their positions and fitness values. Moreover, it randomly selects a portion of individuals as leaders and the remaining individuals as followers. Individual positions are adjusted using corresponding update formulas based on different roles. When there is a local optimum or environmental change, the alert mechanism is triggered to make the population jump out of the local optimum. This process is repeated until the max amount of iterations is reached or the fitness value meets the stopping condition [17]. The optimal position in the population is taken as the optimal solution to the problem. SSA is mainly divided into three types of roles, namely leader, follower, and alert. Leaders are responsible for leading the group to search for food, mainly conducting global searches, and have strong exploratory abilities. Followers follow the leader in the group, mainly conducting local searches to approach the optimal solution. When there is danger or food scarcity in the environment, the alert personnel will issue warning signals, causing the group to adjust their search strategy and jump out of local optimal solutions. The leader is responsible for global search, and the location update formula is shown in Eq. (6).

$$\begin{cases} X_i^{t+1} = X_i^t \cdot \exp\left(-\frac{2r_1}{L}\right), & r_1 < ST \\ X_i^{t+1} = X_i^t + \beta \cdot (X_i^t - X_w), & r_1 \geq ST \end{cases} \quad (6)$$

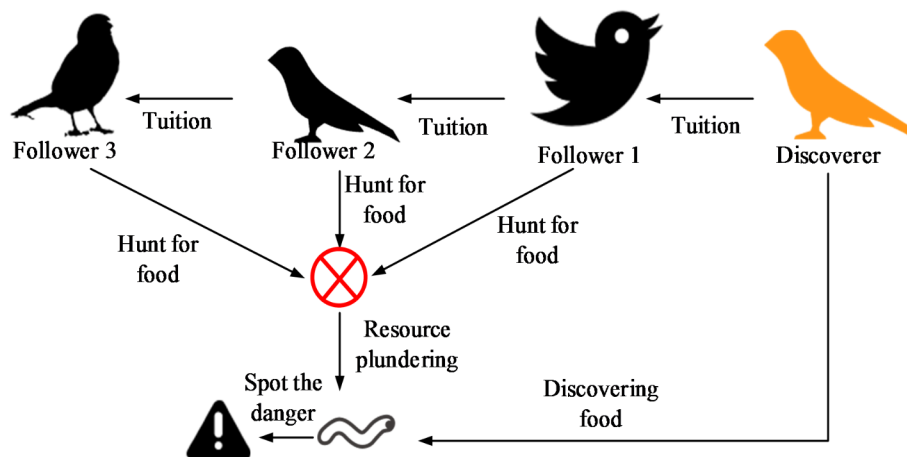


Fig. 4 SSA structure diagram

In Eq. (6), X_i^t represents the current position of the i th sparrow individual at the i th iteration. r_1 denotes a random number with a value range of 0 to 1. ST represents the warning threshold, β represents the coefficient of the control step size, and X_w denotes the position of the current optimal solution in the population. Followers mainly perform local searches, and their position update formula is indicated in Eq. (7).

$$X_i^{t+1} = X_p^t + |X_i^t - X_p^t| \cdot \log\left(\frac{1}{r_2}\right), \quad r_2 \neq 0 \quad (7)$$

In Eq. (7), X_p^t represents the leader with the best position in the population, and r_2 represents a random number with a value range of 0 to 1. When there is danger in the group environment, the alert personnel will adjust their position, and the formula is shown in Eq. (8).

$$X_i^{t+1} = X_i^t + s \cdot (X_i^t - X_{mean}) \quad (8)$$

In Eq. (8), S represents the step size factor used to control the adjustment amplitude, and X_{mean} represents the average position of the population. However, sparrow populations tend to converge too quickly during the foraging, meaning that the algorithm focuses too much on local search in the early stages of iteration, quickly finding a local optimal solution but unable to continue exploring a better global optimal solution. This phenomenon may cause the algorithm to fall into local optima, affecting the final optimization effect. Therefore, the study improves it by using FA, which simulates the luminescent attraction behavior of fireflies at night to search. Fireflies attract other fireflies to approach based on brightness, which is related to the objective function value. The higher the brightness of the firefly, the better the solution, and other fireflies will be attracted closer to the firefly with higher brightness, thus achieving global search [18]. In standard SSA, individual position updates mainly rely on the behavior of leaders and followers. By introducing the attraction mechanism of the SSA, dynamic adjustments can be made based on individual fitness values, as expressed in Eq. (9).

$$X_i^{t+1} = X_i^t + \beta \cdot \exp(-\gamma \cdot r_{ij}^2) \cdot (X_j^t - X_i^t) + \alpha \cdot (rand - 0.5) \quad (9)$$

In Eq. (9), X_i^t and X_j^t represent the positions of the i th and j th sparrow individuals, respectively, and β represents the attraction coefficient, which controls the attraction strength between individuals. γ represents the attenuation coefficient of light intensity, which controls the influence of distance on attraction. r_{ij} means the distance between the i th and j th sparrow individuals, and α represents the disturbance coefficient. The FA-SSA model process is shown in Fig. 5.

In Fig. 5, first is to set the parameters, including population size, maximum iteration times, etc. Next is to initialize the population position, calculate the fitness values of each individual, and sort them. Subsequently, the positions of discoverers in the population are updated, and the positions of early warning individuals are updated to enhance local development capabilities. Meanwhile, the firefly optimization mechanism is utilized to adjust the position of newly added individuals, and combined with firefly dynamic strategies to update the position of other individuals in the population, in order to increase global exploration capability. Then, it should determine whether the iteration termination condition is met. If the conditions are not met, continue iterating and

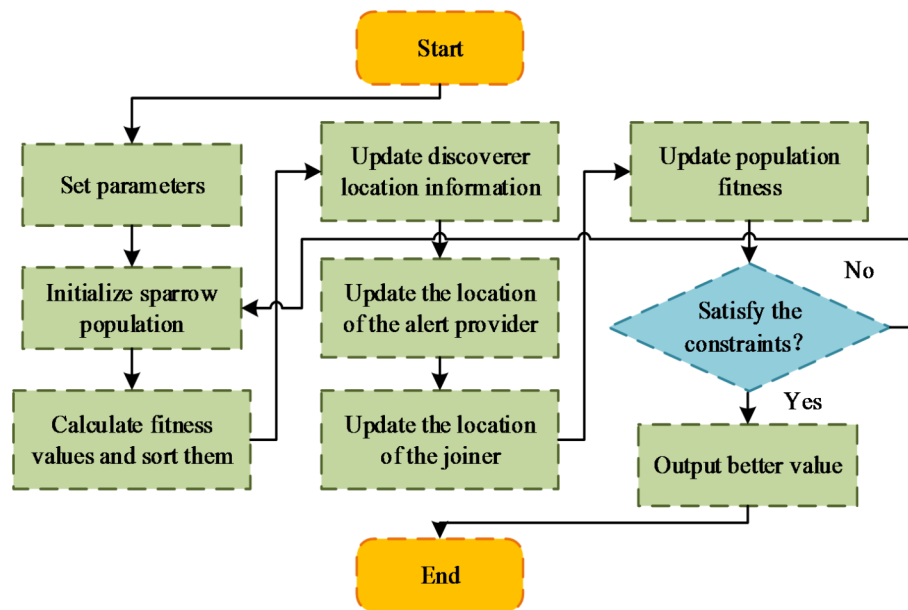


Fig. 5 FA-SSA model process

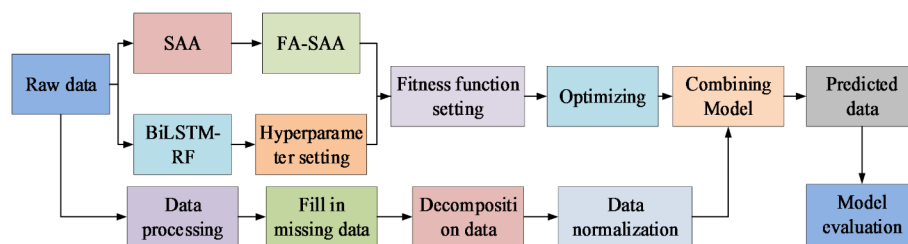


Fig. 6 FA-SSA-BiLSTM-RF combination prediction model

updating. Otherwise, output the optimal solution and end the process. The parameters of BiLSTM-RF are optimized using FA-SSA, and the model structure is shown in Fig. 6.

As shown in Fig. 6, firstly, the model starts from the raw data and preprocesses the data through abnormal data processing and missing data filling steps. The data is then decomposed into multiple sub-models to improve the stability of the data. After completing data preprocessing, data normalisation is performed to ensure that all features are within the same dimension. Next, FA-SSA is used to optimize the BiLSTM-RF model. Firstly, standard SSA and FA-SSA are used to find the fitness function and set the optimal hyperparameter combination to improve the performance of the prediction model. Then, an FA-SSA-BiLSTM-RF prediction model is constructed to predict the sub modal data separately, and the prediction results are subjected to inverse normalization processing to synthesize the final prediction results. Finally, the ACC of the model is evaluated and the predictions are output to complete the process. The pseudocode of the proposed FA-SSA-BiLSTM-RF prediction model is shown in Fig. 7.

FA-SSA-BiLSTM-RF Combined Prediction Model Pseudo Code

```

# Initialize parameters
population_size = 50 # Population size
max_iterations = 100 # Maximum number of iterations
fitness_function = define_fitness_function() # Define the fitness function
best_solution = None # Best solution
best_fitness = inf # Best fitness value

# Initialize population
population = initialize_population(population_size)

# Iterative optimization process
for iteration in range(max_iterations):
    # Calculate fitness and sort population
    fitness_values = [calculate_fitness(individual) for individual in population]
    sorted_population = sort_population_by_fitness(population, fitness_values)

    # Update explorer positions
    for i in range(len(population)):
        update_explorer_position(population[i], fitness_values[i])

    # Update warmer positions (enhance local exploration ability)
    for i in range(len(population)):
        update_warmer_position(population[i], fitness_values[i])

    # Introduce firefly optimization mechanism
    for i in range(len(population)):
        update_firefly_position(population[i], fitness_values[i])

    # Update other individuals' positions (increase global exploration ability)
    for i in range(len(population)):
        update_other_positions(population[i])

    # Termination condition check
    if check_termination_condition(fitness_values):
        break

    # Update best solution
    current_best = get_best_solution(population, fitness_values)
    if current_best['fitness'] < best_fitness:
        best_solution = current_best
        best_fitness = current_best['fitness']

# After optimization, use FA-SSA-BiLSTM-RF for prediction
# 1. Data preprocessing: Handle anomalies, fill missing values
preprocessed_data = preprocess_data(raw_data)

# 2. Decompose data into sub-modes
submodes_data = decompose_data(preprocessed_data)

# 3. Normalize data
normalized_data = normalize_data(submodes_data)

# 4. Optimize BiLSTM-RF model using FA-SSA
optimized_parameters = optimize_bilstm_rf_model(normalized_data, best_solution)

# 5. Make predictions
predictions = predict_with_bilstm_rf(optimized_parameters, normalized_data)

# 6. Inverse normalization of prediction results
final_predictions = inverse_normalize(predictions)

# 7. Model accuracy evaluation
model_accuracy = evaluate_model(final_predictions, true_labels)

# Output the final results
print("Best solution:", best_solution)
print("Model prediction accuracy:", model_accuracy)

```

Fig. 7 FA-SSA-BiLSTM-RF combined prediction model pseudo code

Results

Performance analysis of wind power generation prediction model based on FA-SSA-BiLSTM-RF model

The hardware configuration applied in this experiment was Intel Core i7-13490KF CPU,

NVIDIA Geforce GTX4060Ti GPU, 8GB VRAM, and 16GB RAM. The dataset adopts the publicly available Kaggle dataset, which contains meteorological information such as wind speed, wind direction, temperature, humidity, pressure, etc. from multiple wind farms, covering multiple hours of time series data. The time range of the dataset is from 2015 to 2020, covering WPG data under different seasons and climate conditions. The data volume is relatively large, containing millions of data records. Each record contains real-time meteorological data and corresponding power generation data for the wind farm. The data types mainly include numerical features such as wind speed, temperature, humidity, wind direction, as well as the target variable power generation. The time granularity of the data is once per hour. In addition, the dataset also includes some processed missing and outlier information, which facilitates researchers in preprocessing and data cleaning. The diversity of the dataset enables it to support multidimensional evaluation of wind power prediction models and has certain practical application value, which can help improve WPG efficiency. 6000 data points are selected and divided into training and validation sets in a 5:1 ratio. BiLSTM-RF model and SSA-BiLSTM-RF model are selected as comparison models, and renamed as Model 1 and Model 2. The FA-SSA-BiLSTM-RF model is named Model 3, and the results are shown in Fig. 8.

Figure 8(a) and (b) showcase the comparison of ACC and F1 scores for each model. According to Fig. 8(a), Model 3 achieved an ACC of 95% after about 30 iterations and stabilized at around 98% after 35 iterations, with the fastest convergence speed and highest ACC. Model 2 converged after 45 iterations, with a final ACC of around 90%, significantly lower than Model 3. Model 1 had the slowest convergence speed, reaching an ACC of 85% only after 35 iterations, and the final ACC was the lowest. In Fig. 8(b), Model 3 also performed the best, achieving an F1 score of 98% after 40 iterations and maintaining stability in subsequent iterations. At 40 iterations, Model 2 achieved an F1 score of 90%, but the stabilization time was relatively late. Model 1 only achieved an F1 score of 85% after 35 iterations, showing the worst performance. The errors of each model were analyzed, and the results are shown in Fig. 9.

Figure 9(a) shows the power load prediction results of each model, and Fig. 9(b) shows the prediction errors of each model. From Fig. 9(a), the predicted results of Model 1 had a large deviation, with significant deviations occurring at 8:00 and 18:00. The load values reached 55 MW and 62 MW respectively, while the Actual Values (AV) were 52 MW and 60 MW, showing obvious underestimation and overestimation. The performance of Model 2 improved, but there were still significant prediction errors around 8:00 and

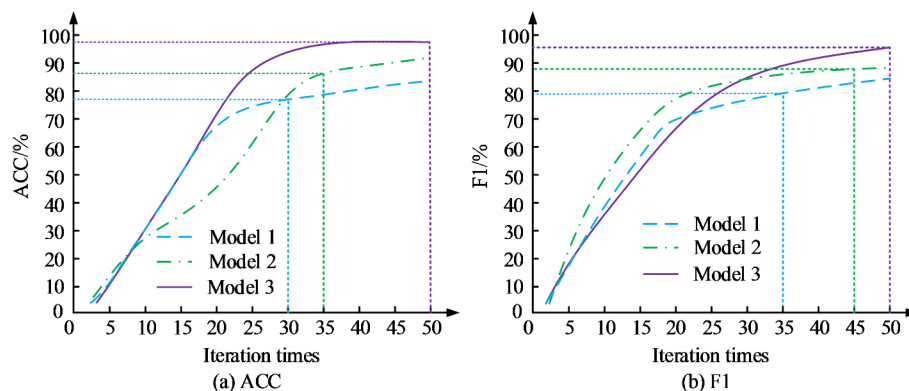


Fig. 8 Comparison of ACC and F1 scores among various models

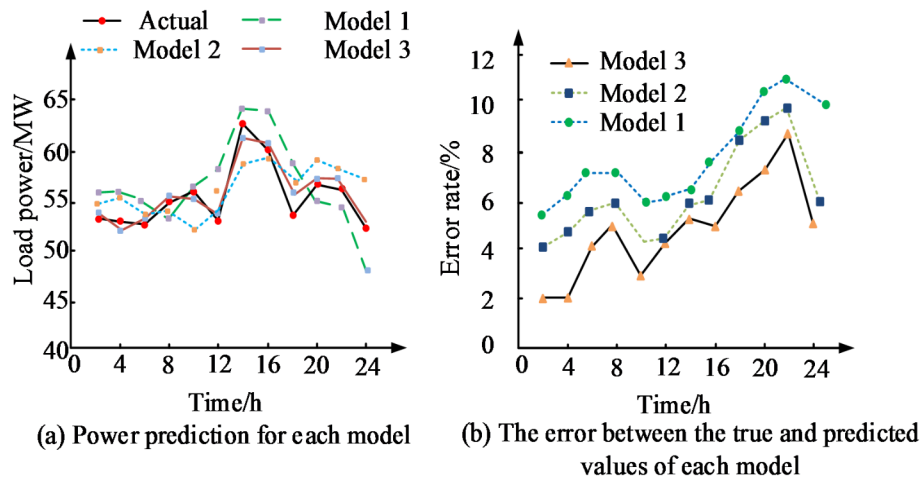


Fig. 9 Error analysis of each model

Table 2 Analysis of ablation test results

Model types	ACC (%)	MSE	MAE	F1 (%)	Average prediction time (s)	Sensitivity to input variations	Computational complexity
BiLSTM-RF	85.2	0.02	0.15	86	0.12	$\pm 2.5\%$	20 MB RAM, 50 ms
SSA-BiLSTM-RF	90.1	0.02	0.12	90.5	0.2	$\pm 3.0\%$	40 MB RAM, 100 ms
FA-BiLSTM-RF	91	0.01	0.12	91.2	0.22	$\pm 2.8\%$	45 MB RAM, 120 ms
FA-SSA-BiLSTM	92.5	0.01	0.11	92.8	0.18	$\pm 2.0\%$	70 MB RAM, 180 ms
FA-SSA-BiLSTM-RF	98	0.01	0.05	98.2	0.25	$\pm 1.5\%$	100 MB RAM, 250 ms

20:00, and the overall trend was similar to the actual load but lacks ACC. Model 3 had the best predictive performance, almost coinciding with the actual load curve, especially near the peak load, with minimal deviation between the predicted and AVs. According to Fig. 9(b), Model 1 had the highest error rate, especially at 8:00 and 18:00, with error rates reaching 10% and 12%, respectively. After incorporating SSA optimization, the error rate of Model 2 decreased slightly, with an average error of around 6%. However, there were still high errors at 12:00 and 20:00. Model 3 had the lowest error rate, with a maximum error of no more than 4%, and the error remained below 2% for most of the time, showing a significant advantage. The experiment findings denoted that Model 3 was greatly better than Model 1 and Model 2, exhibiting higher prediction ACC and lower error rates. Using ablation experiments, the effects of each module in the model were analyzed, and the outcomes are denoted in Table 2.

According to Table 2, BiLSTM-RF performed the worst. SSA-BiLSTM-RF significantly improved model performance by introducing SSA, with ACC increasing by nearly 5% and F1 score increasing by 4.5%. FA-BiLSTM-RF was optimized using FA, which performed better than SSA, mainly because FA has stronger global search capabilities. After removing RF, the performance of FA-SSA-BiLSTM decreased, indicating that RFs play an important role in feature selection and nonlinear modeling. FA-SSA-BiLSTM-RF achieved the best performance, with ACC reaching 98%, indicating that the joint optimization of FA and SSA, as well as the nonlinear modeling ability of RFs, effectively improved the overall model performance. In sensitivity analysis, as the complexity of the model increased, the response of sensitivity to input changes gradually became stronger.

The sensitivity variation of BiLSTM-RF was $\pm 2.5\%$, indicating that the model is relatively stable to changes in input data, while FA-SSA-BiLSTM-RF had the lowest sensitivity, only $\pm 1.5\%$, indicating its higher stability under different input conditions. In terms of computational complexity, as the complexity of the model increased, memory usage and prediction time gradually increased. BiLSTM-RF had lower memory and computational requirements, with only 20 MB RAM and 50 ms prediction time, while FA-SSA-BiLSTM-RF required 100 MB RAM and 250 ms prediction time, indicating that the performance advantage of this model requires more computing resources. The ablation test showed that the joint optimization strategy of FA and SSA, as well as the nonlinear modeling of RFs, significantly improved the performance of the model. Although the prediction time of FA-SSA-BiLSTM-RF is slightly longer, in practical applications, the high ACC and reliability brought by this performance improvement far outweigh the time overhead.

Simulation experiment testing

To further validate the performance of the model, simulation analysis was conducted by selecting one-year WPG data from two areas. The results are shown in Fig. 10.

Figure 10(a) shows the performance of power load forecasting for area A from January to December, and Fig. 10(b) shows the results of power load forecasting for area B from January to December. According to Fig. 10(a), in January and July, the actual load values were 50 MW and 45 MW, respectively, while the model PVs were 48 MW and 47 MW, indicating that the PVs were essentially aligned with the AVs and have a small error. May to August was the peak load period, and the prediction error slightly increased. Especially in August, the PV was 43 MW, while the AV was 40 MW, showing a slight overestimation. Overall, Model 3's predicted performance in area A was relatively stable with small errors. As shown in Fig. 10(b), in July and August, the actual loads were 30 MW and 35 MW, respectively, while the PVs were 28 MW and 32 MW, slightly lower than the AVs, indicating a slight underestimation. The period from October to December was a period of load decline, with actual loads below 15 MW. The PVs of the model almost coincided with the AVs, and the error was small. The load fluctuation in area B was significant, especially in summer and winter, but Model 3 could still capture the trend of load changes well. The prediction time of each model was compared, and the findings are indicated in Fig. 11.

Figure 11(a) and (b) show the simulation time for each model in areas A and B, respectively. According to Fig. 11(a), during the periods of 0–3 months and 4–6 months, Model 1 had the highest prediction time, close to 0.35 s, Model 2 had a time of around 0.3 s,

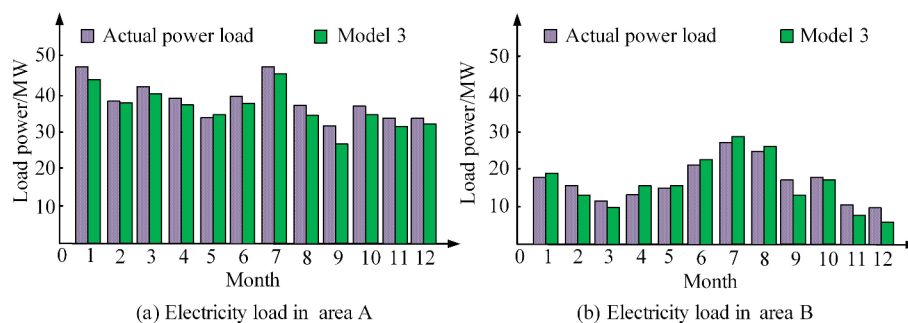


Fig. 10 Wind power generation prediction and actual power generation results of model 3 in regions A and B

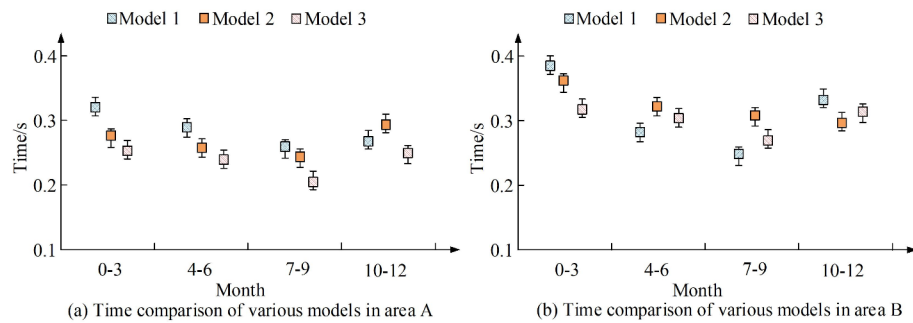


Fig. 11 Comparison of simulation time for Model 1, Model 2, and Model 3 in different quarters

Table 3 Simulation analysis

Area	Index	Model 1	Model 2	Model 3	Area	Index	Model 1	Model 2	Model 3
Area A	ACC/%	86.50	91.20	98.50	Area B	ACC/%	84.00	89.50	97.80
	MSE	0.02	0.01	0.01		MSE	0.02	0.02	0.01
	MAE	0.15	0.11	0.05		MAE	0.16	0.13	0.06
	F1/%	87.00	92.00	98.80		F1/%	85.50	91.00	98.00
	Time/s	0.13	0.21	0.18		Time/s	0.14	0.23	0.19
	Iterations	30	20	12		Iterations	35	22	15

and Model 3 had the shortest time, only 0.25 s. In 7–9 months and 10–12 months, the prediction time of all models slightly decreased, especially Model 3, which remained below 0.2 s. The time for Model 1 and Model 2 was relatively stable, but still slightly longer than Model 3. In Fig. 11(b), similar to area A, Model 1 had the longest time, reaching around 0.4 s, Model 2 had a slightly shorter time of around 0.3 s, and Model 3 had the shortest time, stabilizing at around 0.2 s. In the months of 4–6 and 7–9, Model 3 showed significant time advantages, especially in 7–9, with a minimum time of 0.15 s, indicating its high efficiency. During the period of 10–12, the prediction time of all models slightly increased, but Model 3 still maintained the lowest time. The findings denoted that the jointly optimized model not only improved ACC, but also significantly reduced computation time. The comprehensive simulation performance of each model was analyzed, and the outcomes are indicated in Table 3.

According to Table 3, Model 3 performed the best in both regions, with ACC reaching 98.5% in area A and 97.8% in area B. This indicated that the joint optimization strategy of FA and SSA could significantly raise the generalization of the model. The computation time of Model 3 was between Model 1 and Model 2, with 0.18 s and 0.19 s respectively in the two regions. It had the fastest convergence speed, requiring only 12 and 15 iterations. The introduction of FA effectively optimized the efficiency of SSA, reducing computation time and iteration times. The experimental results showed that Model 3 performed the best in both area A and area B, significantly outperforming Model 1 and Model 2 in ACC, error, computational efficiency, and robustness. The joint optimization strategy of FA and SSA significantly improved the performance of the model, making it more accurate and stable in handling complex power load forecasting tasks. Different algorithms were introduced to compare their performance, and the results are shown in Table 4.

According to Table 4, the SSA-FA-BiLSTM-RF model performed the best in all evaluation metrics, with an ACC of 98.5%, far higher than BiLSTM-RF's 92.1% and LSTM's

Table 4 Performance analysis of different algorithm models

Model	ACC/%	MSE	RMSE	MAE	Time/s
SSA-FA-BiLSTM-RF	98.5	0.005	0.070	0.050	0.18
SSA-BiLSTM-RF	95.3	0.009	0.095	0.075	0.22
FA-BiLSTM-RF	94.8	0.011	0.105	0.082	0.25
BiLSTM-RF	92.1	0.015	0.122	0.097	0.300
LSTM	89.7	0.018	0.134	0.112	0.35
GRU	88.5	0.021	0.145	0.126	0.38
RNN	86.2	0.025	0.158	0.135	0.42

89.7%. This indicates that the joint optimization of SSA and FA has significant advantages in improving prediction ACC. In terms of MSE, the SSA-FA-BiLSTM-RF model had the lowest MSE, only 0.005, far lower than BiLSTM-RF's 0.015 and LSTM's 0.018, indicating that the proposed model is superior to other comparative models in error control. The RMSE and MAE indicators further validated the advantages of the model, with an RMSE of only 0.07 and an MAE of only 0.05, significantly better than the RMSE of 0.122 and MAE of 0.097 for BiLSTM-RF and 0.134 and MAE of 0.112 for LSTM. This indicates that SSA-FA optimization can effectively reduce prediction bias and improve model stability. In addition, in terms of computational efficiency, the SSA-FA-BiLSTM-RF model had a prediction time of only 0.18 s, which was better than SSA-BiLSTM-RF's 0.22 s, FA-BiLSTM-RF's 0.25 s, and BiLSTM-RF's 0.30 s. This indicates that SSA-FA optimization not only improves prediction ACC, but also reduces computation time to a certain extent.

At present, WPG forecasting still faces many challenges. Firstly, the high randomness of environmental factors such as wind speed and meteorological conditions makes it difficult for prediction models to maintain stable ACC under all conditions. How to further improve the adaptability of the model under extreme weather conditions remains a challenge. Secondly, there is still room for optimization in the computational efficiency of the current model, especially in terms of real-time prediction ability on large-scale datasets, which needs to be further improved to meet the needs of power grid scheduling. In addition, wind power forecasting not only involves short-term load forecasting, but also needs to consider long-term trend analysis. In the future, the robustness of forecasting can be improved by combining multi-scale time series modeling methods. In terms of model optimization, it can further explore adaptive hyperparameter adjustment mechanisms to reduce the manual setting of parameters and improve the generalization ability of the model. In addition, combining multimodal data such as satellite remote sensing data, meteorological radar data, etc., may further improve prediction ACC and stability. At the application level, how to apply optimized WPG prediction models to smart grids and achieve dynamic load regulation will be an important direction for future research.

Conclusion

A BiLSTM-RF combined model based on FA and SSA joint optimization was developed to address the high randomness and nonlinear modeling problems in WPG prediction. This model extracted time series features through BiLSTM, used RF for feature selection and nonlinear modeling, and employed FA-SSA joint algorithm for hyperparameter optimization. The experimental results showed that FA-SSA-BiLSTM-RF achieved an ACC of 95% after about 30 iterations and stabilized at around 98% after 35 iterations, with the fastest convergence speed and highest ACC. The ACC of the FA-SSA-BiLSTM-RF model

in area A and area B reached 98.5% and 97.8%, respectively, with MAEs reduced to 0.005 and 0.006, respectively. Compared to the traditional BiLSTM-RF model, the optimized model reduced computation time by 20% and improved convergence speed by 50%. In July and August, the actual loads were 30 MW and 35 MW respectively, while the PVs were 28 MW and 32 MW. The PVs were slightly lower than the AVs, indicating a slight underestimation. The period from October to December was a period of load decline, with actual loads below 15 MW. The PVs of the model almost coincided with the AVs, and the error was small. This indicates that the performance of the model proposed by the research is significantly superior and can be applied to the field of WPG prediction. However, there were still shortcomings in the research, and the model still had certain errors in handling load forecasting under extreme weather conditions. Future research can further explore the addition of environmental factors and the improvement of model structure to enhance the adaptability and stability of the model.

Attachment

Symbol	WPG	BiLSTM	LSTM	RF	SSA	FA	ACC
Definition	Wind Power Generation	Bidirectional Long Short-Term Memory	Long Short-Term Memory	Random Forest	Sparrow Search Algorithm	Firefly Algorithm	Accuracy
Symbol	RMSE	MAE	F1	PV	AV	Time	Nodes
Definition	Root Mean Squared Error	Mean Absolute Error	F1 score	Predicted Value	Actual Value	Prediction Time	Number of Nodes

Author contributions

WZ and HY provided the concept and wrote the draft; LX and LS revised this paper critically. All the authors reviewed this paper carefully and approved this submission.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

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An ethics statement was not required for this study type, no human or animal subjects or materials were used.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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